

Canonical bases and coherent sheaves

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1) Kazhdan-Lusztig theory

$\mathfrak{g}$ ss Lie alg / $\mathbb{C}$, $M_\lambda = U\mathfrak{g} \otimes_{U\mathfrak{b}} \mathbb{C}_\lambda$ - Verma module

$\downarrow$

L_λ irred quotient

Problem: compute char of L_λ

Sol'n (KL et al.) Consider the cat. generated by L_λ (cat $\mathcal{O}$)

Focus on principal block $\mathcal{O}_0 \ni \mathbb{C}$, irred. $L_{w,0}$ (cartan action diagonalizable)

$$K(\mathcal{O}_0) \simeq \mathbb{Z}[w]$$

$$[M_{w,(-2\rho)}] \mapsto w$$

Then $[L_{w,(-2\rho)}] \mapsto C_w$ - canonical basis.

Key features

- relation to geometry

Localization theorem: $\mathcal{O}_0 \simeq D\text{-mod}_B(\mathfrak{g}/u) \simeq \text{Perf}_B(\mathfrak{g}/u)$

- HC bimodules description

$$D\text{-mod}_B(\mathfrak{g}/u) \simeq D\text{-mod}_\mathfrak{g}(\mathfrak{g}/B \times \mathfrak{g}/u) \simeq (\mathfrak{g} \oplus \mathfrak{g}, \mathfrak{g})\text{-mod}^{0,\hat{0}}$$

- Symmetries 1) $D^b(\mathcal{O}_0) \supset B \times B$

B - Artin braid group

comes from correspondences

$$\begin{array}{ccc} & (\mathfrak{g}/B)^2_w & \\ \swarrow & & \searrow \\ \mathfrak{g}/B & & \mathfrak{g}/B \end{array}$$

ii) Wall crossing functor

$$\mathbb{E}_\alpha \triangleq T_{\mu \rightarrow 0} \circ T_{0 \rightarrow \mu} \quad \mu \text{ on the } \alpha \text{ wall}$$

T-translation functor

$\mathbb{E}_\alpha: \mathcal{O}_0 \rightarrow \mathcal{O}_0$ - exact self-adjoint, sends projective to projective

iii) Zuckerman functors

$Z_\alpha: D^b(\mathcal{O}_0) \rightarrow D^b(\mathcal{O}_0)$ comes from $\pi_\alpha^* \pi_{\alpha*}$, $\pi_\alpha: \mathcal{G}/B \xrightarrow{\mathbb{P}^1} \mathcal{G}/P_\alpha$

Z_α - semisimple (deep fact $\Rightarrow$ KL conj.)

VB irred Lw.o are (up to shift) summands of $Z_{\alpha_1} \dots Z_{\alpha_n} (L_{-2\rho})$
 $\parallel$
 $M_{-2\rho}$

Indecomposable projectives are summands of $\mathbb{E}_{\alpha_1} \dots \mathbb{E}_{\alpha_n} (M_0)$

Their classes give canonical basis c_w and dual canonical basis c_w' .

Also summands of $\mathbb{E}_{\alpha_1} \dots \mathbb{E}_{\alpha_n} (L_{-2\rho} = M_{-2\rho})$ are tilting objects, give twisted dual canonical basis.

- Koszul duality [BGS] is an equiv. $D^b(\mathcal{O}_0^{\text{ev}}) \simeq D^b(\mathcal{O}_0^{\text{od}})$

interchanges irred. & projectives

(variant irred & tilting)
 $\underbrace{\hspace{10em}}$

Coherent sheaves

$$\tilde{V} = T^*(\mathcal{G}/B)$$

canonical basis in $k(\text{Ch}(\tilde{V}))$ controls characters of tilting modules for quantum groups at $q = q_1$,

also the objects representing elements of the basis contain information about $H^*(b_q, T_\lambda), H^*(u_q, T_\lambda)$, T_λ - tilting, b_q - Brel in the small quantum group u_q .

Also controls characters of some $\hat{\mathfrak{g}}$ -modules.

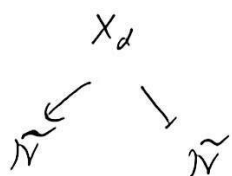


$D^b(\text{coh}^q(\tilde{N})) \hookrightarrow B_{\text{aff}}$ - extended affine braid group

$$B_{\text{aff}} = \langle \tilde{s}_\alpha, \theta_\lambda : \lambda \in \Lambda, \alpha \in \Sigma \rangle \quad \begin{array}{l} \Lambda - \text{weight lattice,} \\ \Sigma = \text{simple roots} \end{array}$$

$$\theta_\lambda: F \mapsto F \otimes \mathcal{O}(\lambda)$$

$\tilde{s}_\alpha$ comes from



$$X_\alpha = \Delta \cup N_\alpha,$$

Δ - diagonal, N_α - conormal to $(G/B)^2_{s_\alpha}$
(up to $\mathcal{O}(-p)$ twist)

$\mathbb{Z}_\alpha, \alpha \in \Sigma_{\text{aff}}$

If $\alpha \in \Sigma$, then (up to a $(-p)$ twist) given by $\begin{array}{ccc} & \overline{N}_\alpha & \\ \swarrow & & \searrow \\ \tilde{N} & & \tilde{N} \end{array}$

For $\alpha_0 \in \Sigma_{\text{aff}} \setminus \Sigma$, $\mathbb{Z}_{\alpha_0} = b \mathbb{Z}_\alpha b^{-1}$ where $\alpha \in \Sigma$, $b \tilde{s}_\alpha b^{-1} = \tilde{s}_{\alpha_0}$.

Th a) Classes of indecomposable summands of $\mathbb{Z}_{\alpha_1} \dots \mathbb{Z}_{\alpha_k}(\mathcal{O})$ are (up to sign) elements of the canonical basis in $K(\text{coh}^q(\tilde{N}))$.

b) These are (up to shifts) irreducibles in the heart of some t -structure

Rem. On the meaning of a) $K(\text{coh}^q(\tilde{N})) \supset W_{\text{aff}}$

$$\begin{aligned} K(\text{coh}^{q \times q}(\tilde{N})) &\supset H_{\text{aff}} - \text{affine Hecke algebra} \\ &\simeq H_{\text{aff}} \otimes_{H_{\text{fin}}} \text{sgn} \leftarrow H_{\text{aff}} \end{aligned}$$

$$\mathbb{Z}[W_{\text{aff}}] \rightarrow K(\text{Coh}^G(\tilde{N}))$$

↓

$$C_w \mapsto 0 \text{ or element of canonical basis.}$$

$w \in W_{\text{aff}}$

What about C_w itself? Dual canonical basis C'_w ?

Answer $\mathbb{Z}[W_{\text{aff}}] \simeq K(\text{Coh}^G(\tilde{N} \times_{\mathfrak{g}} \tilde{N}))$

Again have $\mathbb{Z}_\alpha, B_{\text{aff}}$ acting on the left & right

Direct summands of $\mathbb{Z}_{\alpha_1} \cdots \mathbb{Z}_{\alpha_n}(\mathcal{O}_\Delta)$ give canonical basis (up to sign)

These are inv'd (up to shift) of some t-structure, related to category $\mathcal{O}$ for $\hat{\mathfrak{g}}$, also to HC bimodules in positive characteristic

NB Need to consider $\tilde{N} \times_{\mathfrak{g}} \tilde{N}$ as a derived scheme.

But also $\mathbb{Z}[W_{\text{aff}}] \simeq K(\text{Coh}^G(\tilde{\mathfrak{g}} \times_{\mathfrak{g}} \tilde{\mathfrak{g}}))$

$$D^b \text{Coh}^G(\tilde{\mathfrak{g}} \times_{\mathfrak{g}} \tilde{\mathfrak{g}}) \supset \mathbb{E}_\alpha$$

$$\tilde{\mathfrak{g}} \xrightarrow{\pi_\alpha} \tilde{\mathfrak{g}}_\alpha \quad \mathbb{E}_\alpha = \pi_\alpha^* \pi_{\alpha*}$$

$$\tilde{\mathfrak{g}}_\alpha = \left\{ (P_\alpha \in \mathfrak{g}/P_\alpha, x \in P_\alpha) \right\}$$

Summands of $\mathbb{E}_{\alpha_1} \cdots \mathbb{E}_{\alpha_n}(\mathcal{O}_\Delta)$ (tilting)

- dual canonical basis C'_w .

Homotopy category of tilting $\text{Ho}(\mathcal{T}) \xrightarrow{\sim} D^b \text{Coh}^G(\tilde{\mathfrak{g}} \times_{\mathfrak{g}} \tilde{\mathfrak{g}}).$

↓

$$\text{Ext}^{>0}(T_1, T_2) = 0, \quad T_1, T_2 \in \mathcal{T}, \quad \langle T_i \rangle = D^b \text{Coh}^G(\tilde{\mathfrak{g}} \times_{\mathfrak{g}} \tilde{\mathfrak{g}}).$$

$$\mathrm{Dolb}^q(\tilde{N} \times_{\tilde{g}} \tilde{N}) \longleftrightarrow \mathrm{Dolb}^q(\tilde{g} \times_{\tilde{g}} \tilde{g})$$

linear Koszul duality
of Mirković-Riche

$$C_W \longleftrightarrow C_W^!$$

Linear Koszul duality generalizes KD between $\Lambda(V)$ and $\mathrm{Sym}(V^*)$

$$\mathrm{Tor}_{\Lambda(V)}^{\mathrm{Sym}(V^*)}(k, k) = \mathcal{O}(\{0\} \times_{V^*} \{0\})$$

$$\mathrm{Ext}_{\Lambda(V)}^{\bullet}(k, k) \simeq \mathrm{Sym}(V^*)$$

$\mathcal{E}$ - vector bundle, $\mathcal{E}_1, \mathcal{E}_2 \subset \mathcal{E}$ subbundles

$\downarrow$
 S

$$\mathrm{Dolb}(\mathcal{E}_1 \times_{\mathcal{E}} \mathcal{E}_2) \longleftrightarrow \mathrm{Dolb}(\mathcal{E}_1^{\perp} \times_{\mathcal{E}^*} \mathcal{E}_2^{\perp})$$

$$\text{ex } S = (G/B)^2, \quad \mathcal{E} = \mathcal{E}^* = g \times (G/B)^2$$

$$\mathcal{E}_1 = \tilde{N} \times_{(G/B)_1} (G/B)^2, \quad \mathcal{E}_2 = \tilde{N} \times_{(G/B)_2} (G/B)^2$$

$$\mathcal{Z}_{\alpha} \longleftrightarrow \mathcal{E}_{\alpha}$$

Let now θ be an involution of G , $K = G^{\theta}$.

$$\text{we have } \tilde{N} \times_{\tilde{g}} k^{\perp} \xrightarrow{\mathrm{LKD}} \bar{g} \times_{\bar{g}} k$$

union of conormals
to K -orbits on G/B
derived scheme

usual (non-derived) scheme

(except if θ is split, i.e. $t^{\theta} = 0$ for a θ -inv (arbit t)

$$D\mathrm{Coh}^k(\tilde{N} \times_{\mathfrak{g}} k^\perp)$$

$\hookrightarrow$

$\mathbb{Z}_2$

$$D\mathrm{Coh}^k(\tilde{\mathfrak{g}} \times_{\mathfrak{g}} k)$$

$\hookrightarrow$

$\mathbb{Z}_2$

Conj. (G-adjoint) summands of $\mathbb{Z}_2, \dots, \mathbb{Z}_2(\mathcal{O}_S) = \mathcal{T}_k$
 preimage of closed orbits

$$H_0(\mathcal{T}) \xrightarrow{\sim} D\mathrm{Coh}^k(\tilde{\mathfrak{g}} \times_{\mathfrak{g}} k) \quad \text{classes of indecomposables}$$

- dual canonical basis.

Rem 1. In the $\tilde{\mathfrak{g}} \times_{\mathfrak{g}} \tilde{\mathfrak{g}}$ case, proofs are based on geometric local Langlands realization of the categories via constructible sheaves on FL-affine flag variety of the dual group.

In the $\tilde{k}$ setting such a realisation is not known, although there're related conjectures by Ben-Zvi - Nadler.

Rem 2 Conj. reduces to $\mathrm{Ext}^{>0}(T_1, T_2) = 0$, $T_1, T_2 \in \mathcal{T}_k$ & $\langle \mathcal{T}_k \rangle = D^b\mathrm{Coh}^k(\tilde{k})$
 (can prove vanishing for some $T_1, T_2 \in \mathcal{T}$).

Rem 3 Category $\mathcal{T}$ should have a Soergel bimodule description.

Recall Soergel bimodules is a full subcat in $\mathrm{Coh}(\mathfrak{t} \times_{\mathfrak{t}/w} \mathfrak{t})$

||

$$\cup \Gamma_w, \quad \Gamma_w \subset \mathfrak{t} \times \mathfrak{t}$$

is the graph of w

$$\mathfrak{t}/w \xrightarrow[k]{\hookrightarrow} \mathfrak{g}^{\mathrm{reg}}, \quad \mathfrak{t} \times_{\mathfrak{t}/w} \mathfrak{t} \subset \tilde{\mathfrak{g}} \times_{\mathfrak{g}} \tilde{\mathfrak{g}}$$

constant slice

In the $(\mathfrak{g}, k)$ -setting, $\tilde{k}_k : \mathfrak{t}_k \rightarrow k$, $\mathfrak{t}_k \times_{\mathfrak{t}/w} \mathfrak{t} \subset \tilde{k} = k \times_{\mathfrak{g}} \tilde{\mathfrak{g}}$

Affine Serre's Bimodule $S_{\text{aff}} [Abe]$ is a full subcat. in

$$\left\{ (M, \gamma) : M \in \text{Gh} \left(t \times_{t/w} t \right), \gamma - W_{\text{aff}} \text{ grading on } M_F = M \otimes \Gamma_{\text{aff}}(k[t]) \right. \\ \left. \text{s.t. } \text{Supp}(M_F) \subset \Gamma_{\bar{w}} \right\}$$

Can define similarly $S_{\text{aff}}^k = \left\{ (M, \gamma) : M \in \text{Gh} \left(t \times_{t/w} t \right) \right\}$
 $\gamma \dots$

Can define a functor $\sigma: \mathcal{T} \rightarrow S_{\text{aff}}$ - an upgrade of $\tilde{\kappa}^*$

$$\sigma_k: \mathcal{T}_k \rightarrow S_{\text{aff}}^k - \text{an upgrade of } \tilde{\kappa}_k^*$$

σ is fully faithful, σ_k is expected to be fully faithful
 on $\mathcal{T}_k$